



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME

CLASS

CHEMISTRY

Preliminary Examination
Paper 3 Free Response

9647/03

27 Aug 2014
2 hours

Additional Materials: Data Booklet
Cover Page
Writing Papers

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **four** questions.
A Data Booklet is provided.
The use of an approved scientific calculator is expected, where appropriate.
You are reminded of the need for good English and clear presentation in your answers

The number of marks is given in the brackets [] at the end of each question or part question.

Start each question on a fresh piece of paper.

At the end of the examination, fasten all your work securely together with the cover page provided and ensure your script is arranged in an orderly manner.

This document consists of **11** printed pages and **1** blank page.

Answer any **four** questions.

- 1 Singapore has hosted the Formula One Grand Prix since 2008. This car racing event sees race car drivers from around the world compete in a circuit to win the top honour of a race car champion.

These race cars are fuelled by petrol, which is a mixture of hydrocarbons. One such hydrocarbon is pure liquid octane, C_8H_{18} .

- (a) (i) Define the term *standard enthalpy change of combustion*.
- (ii) Use the data in the table below to calculate the standard enthalpy change of combustion of octane. Hence calculate the heat evolved per gram of octane.

species	$\Delta H_f / \text{kJ mol}^{-1}$
$\text{C}_8\text{H}_{18}(\text{l})$	-269
$\text{CO}_2(\text{g})$	-394
$\text{H}_2\text{O}(\text{l})$	-286

- (iii) Methanol, CH_3OH , is also commonly used as a fuel for high-performance engines in race cars. Due to the high temperature in car engines, methanol is converted to the vapour state.

Use the bond energies given in the *Data Booklet* to calculate the standard enthalpy change of combustion of methanol.

Hence calculate the heat evolved per gram of methanol.

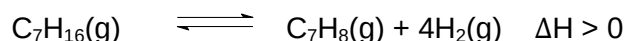
- (iv) Using your answers in (a)(ii) and (a)(iii), suggest an advantage of using liquid octane over methanol as the fuel in race cars.

[6]

- (b) Methylbenzene is a component of lead-free petrol. It is synthesised by cyclising and dehydrogenating heptane.

Methylbenzene can be made by catalytic reforming of heptane. This process causes straight chain hydrocarbons between 6 to 8 carbon atoms to rearrange into compounds containing benzene rings.

This process requires a platinum catalyst, high temperature of 500 °C and pressure of about 20 atm.



- (i) State *Le Chatelier's Principle*.
- (ii) Suggest reasons why these operating conditions are used.

[4]

- (c) Compound **P** has the molecular formula $\text{C}_5\text{H}_9\text{ON}$.

- **P** exhibits optical isomerism, is neutral, and reacts with sodium metal.
- On reacting **P** with lithium aluminium hydride, compound **Q**, $\text{C}_5\text{H}_{13}\text{ON}$, is formed.
- When **Q** is refluxed with aqueous acidified potassium manganate(VII), **R**, $\text{C}_5\text{H}_{11}\text{O}_2\text{N}$ is obtained.
- The addition of thionyl chloride, SOCl_2 to **R** produces **S**.
- Under suitable conditions, **S** forms **T**, $\text{C}_5\text{H}_9\text{ON}$, which is neutral, along with white fumes of HCl .

Explain and identify the structures of **P**, **Q**, **R**, **S** and **T**.

[10]

[Total: 20]

- 2 Iron forms compounds mainly in the +2 and +3 oxidation states. Iron compounds are coloured and were widely used in paintings during the 19th century.

- (a) Explain why iron exhibits variable oxidation states but not calcium. [2]
- (b) Explain why most iron compounds are coloured. [3]
- (c) When Vincent van Gogh painted the “The Starry Night” in 1889, he used a lot of the pigment known as Prussian Blue, which is an iron complex, $\text{Fe}_4(\text{Fe}(\text{CN})_6)_3$.

Draw the structure of the complex ion in Prussian Blue, stating the oxidation number of iron in the complex ion. [2]

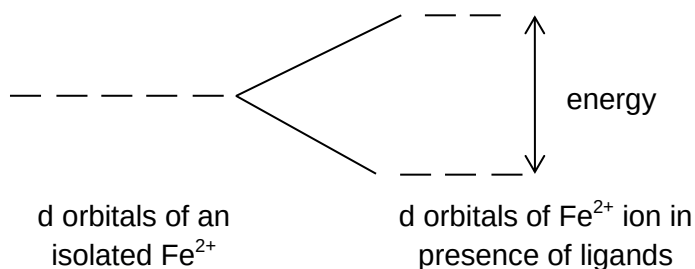
- (d) When water ligands in a hydrated metal ion are substituted by other ligands, the equilibrium constant for the reaction is referred to as the stability constant, K_{stab} of the new complex.



The following table lists two stability constants for the above reaction. (where n is a whole number)

L	n	K_{stab}
SCN^-	1	9×10^2
CN^-	6	1×10^{34}

- (i) Write the expression for the stability constant, K_{stab} , for $\text{L} = \text{SCN}^-$.
- (ii) Use the data given in the table to predict and explain which is the predominant complex formed when a solution containing equal concentrations of both SCN^- and CN^- ions was added to a solution containing $\text{Fe}^{3+}(\text{aq})$ ions. [3]
- (e) The following diagram shows how the d orbitals are split in an octahedral environment.



The arrangement of electrons in the d orbitals depends on the spin states of complexes. In a ‘high spin’ state, the electrons occupy all the d-orbitals singly, before starting to pair up in the lower energy d-orbitals.

In a ‘low spin’ state, the lower energy d orbitals are filled first, by pairing up if necessary, before the higher energy d orbitals are used.

- (i) Use diagrams, like the one on page 4, to show the electronic configuration of a Fe^{3+} ion in a high spin state and in the low spin state.

Show in your diagram, the relative size of the energy gap, ΔE , for each spin state.

Many transition elements and their compounds are paramagnetic, which means that they are attracted to a magnetic field. This property is due to the presence of unpaired electrons in the d orbitals.

Relative paramagnetism is directly proportional to the number of unpaired electrons in the d orbitals.

The table below shows the relative paramagnetism of some iron complexes.

formula of complex	relative paramagnetism
$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	5
$[\text{Fe}(\text{CN})_6]^{3-}$	1

- (ii) Using your answer to (e)(i) and the information given above concerning paramagnetism of the different complexes, state and explain which ligand, H_2O or CN^- , will cause a larger energy gap, ΔE , between its d orbitals.
- (iii) Deduce the relative paramagnetism of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$, given that a solution containing $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is pink in colour.

[6]

- (f) Siderite is a mineral composed of iron(II) carbonate. It is a valuable iron mineral, since it is 48% by mass of iron and contains no sulfur or phosphorus.

Limestone which compose of calcium carbonate, is a common substance found in rocks and shells.

- (i) Explain, with reasoning, why calcium carbonate has a higher decomposition temperature than iron(II) carbonate.
- (ii) Explain, with reasoning, whether the third ionisation energy of Ca is larger or smaller than that of iron.

[4]

[Total: 20]

- 3 Magnesium and its compounds are used as commercial products and in industrial applications. Lithium compounds resemble those of magnesium due to their diagonal relationship in the Periodic Table.

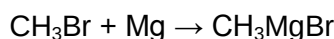
- (a) (i) In aqueous solutions, both magnesium and lithium cations exist as $[M(H_2O)_n]^{y+}$, where n is the solvation number and y is its overall charge.

Construct a balanced equation to illustrate the behaviour of magnesium ion in water. Hence, suggest a pH value of the resulting solution.

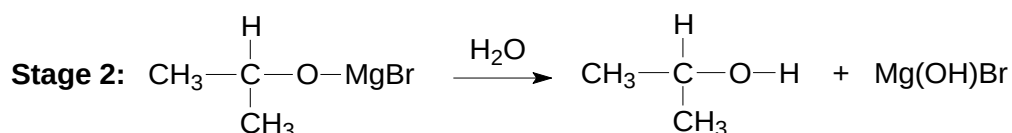
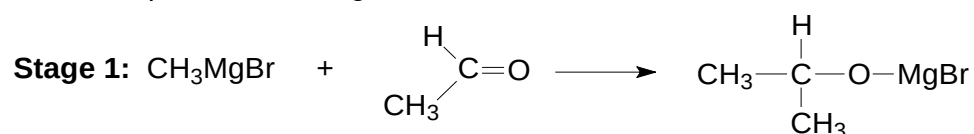
- (ii) Despite having similar charge density, the solvation number, n , is 4 for Li^+ . Suggest one reason for this.

[3]

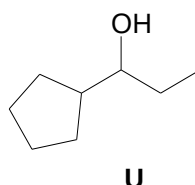
- (b) Magnesium forms an important group of covalent compounds known as Grignard reagents. In preparation of Grignard reagents, a halogenoalkane is added to small pieces of magnesium in a flask containing dry ether as solvent. The reaction is as follows:



This Grignard reagent can be used to synthesise propan-2-ol from ethanal. The reaction takes place in two stages as shown below:



- (i) State the type of reaction taking place in **Stage 1**.
- (ii) Organolithium compounds are synthesised in the similar way as the Grignard reagent described above. An additional lithium salt is produced as by-product.
Write an equation to show how the lithium alkyl is formed when bromoethane is added to lithium metal.
- (iii) Compound **U**, shown below, may be synthesised using the lithium alkyl produced in (b)(ii) reaction.



Suggest a suitable carbonyl compound that can be used to synthesise compound **U**.

[3]

- (c) In the production of magnesium from seawater, magnesium is precipitated as its hydroxide and converted to its chloride by reaction with hydrochloric acid.

Magnesium chloride is then recovered by evaporation of its aqueous solution. Finally, magnesium metal is obtained by electrolysis of the molten salt.

- (i) During the electrolysis of molten magnesium chloride using inert electrodes, a current of 8.0 A was passed through the cell for 67 hours.

Calculate the mass of magnesium extracted.

- (ii) Chlorine gas produced at the anode is then pumped into reinforced cylinders each with a capacity of 2.0 dm^3 . All cylinders are maintained at a pressure of 1100 kPa and 25°C .

Determine the amount of chlorine gas produced and hence the number of cylinders needed to store the chlorine gas produced in this duration.

- (iii) In a similar electrolytic set-up involving magnesium chloride, oxygen gas and hydrogen gas were produced instead.

Construct an overall equation for this electrolysis.

Hence, draw a fully labelled diagram of the electrolytic cell you would set up to collect these gases under standard conditions. Show clearly the relative volume of gases on your diagram.

[9]

- (d) Magnesium phosphate, $\text{Mg}_3(\text{PO}_4)_2$, is a sparingly soluble salt which is used as laxatives and antacids.

(K_{sp} of magnesium phosphate at $25^\circ\text{C} = 1.04 \times 10^{-24} \text{ mol}^5 \text{ dm}^{-15}$)

- (i) Calculate the solubility of magnesium phosphate in water at 25°C .
- (ii) A saturated solution containing magnesium hydroxide, $\text{Mg}(\text{OH})_2$, and magnesium phosphate has a pH of 7.5.

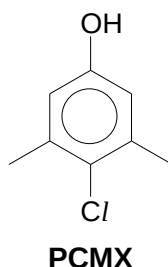
Given that the K_{sp} of magnesium hydroxide is $2.00 \times 10^{-13} \text{ mol}^3 \text{ dm}^{-9}$, calculate the concentration of phosphate, and hence the solubility of magnesium phosphate in this saturated solution.

[5]

[Total: 20]

- 4 Before the mid 1800s, major surgery was often fatal due to post-operative infections. This changed in the 1860s when Joseph Lister developed antiseptic surgery using phenol to sterilise wounds and instruments.

In fact, most phenolic derivatives have antiseptic properties. The active germ-killing ingredient in *Dettol*® is 4-chloro-3,5-dimethylphenol, also known as PCMX.



- (a) The pK_a values of weak acids are given below.

- PCMX $pK_a = 9.7$
- phenol $pK_a = 9.9$
- ethanol $pK_a = 15.9$

Account for the difference in acidity of the acids.

[4]

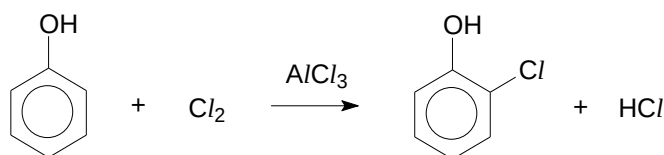
- (b) Given that the solubility of PCMX is 330 mg dm^{-3} , use relevant information from (a) to calculate the pH of a saturated solution of PCMX.

[2]

- (c) Account for the low solubility of PCMX in water in terms of chemical structure and bonding.

[2]

- (d) (i) Chlorination of phenol is represented by the following overall equation:



The Fridel craft alkylation of phenol proceeds via a similar mechanism with halogenoalkane where $AlCl_3$ reacts in a similar way with halogenoalkanes, RX , producing a carbocation that can then attack the benzene ring.

Describe the mechanism involved when phenol reacts with CH_3Cl .

- (ii) Suggest why PCMX cannot be synthesised from Fridel crafts alkylation of 4-chlorophenol.

[5]

Weak organic acids are also known to have some mild antiseptic effects. The table below compares the pK_a values of oxalic acid with ethanoic acid.

	Formula	pK_{a1}	pK_{a2}
Ethanoic acid	CH_3COOH	4.76	-
Oxalic acid	HO_2CCO_2H	1.19	4.21

- (e) (i) Suggest why the pK_{a1} of oxalic acid is much lesser than the pK_{a1} of ethanoic acid.
- (ii) Suggest why the pK_{a2} value of oxalic acid is higher than the pK_{a1} of oxalic acid.
- (iii) 30 cm^3 of 0.10 mol dm^{-3} of NaOH (aq) is added to 10 cm^3 of 0.10 mol dm^{-3} of oxalic acid in a conical flask.

Calculate the initial pH (ignore effect of pK_{a2}) of oxalic acid and the final pH after adding 30 cm^3 of NaOH (aq).

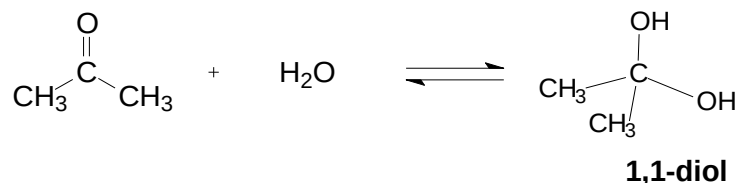
Sketch a clearly labelled the titration curve:

- Initial pH and final pH
- Equivalence volume(s)
- Circle any buffer region(s) and corresponding pH at maximum buffering capacity

[7]

[Total:20]

- 5 Propanone undergoes reaction with water to yield 1,1,-diols. This hydration reaction is reversible.



- (a) (i) In the base-catalysed reaction, the attacking species is the hydroxide ion. Outline the two step mechanism of the reaction of the base-catalysed reaction of propanone with water.

- (ii) The relationship between $\Delta G^\ominus$ and K_c for the reaction between propanone and water is

$$\Delta G^\ominus = -RT \ln K_c$$

where $\Delta G^\ominus$ is the standard Gibbs free energy change in J mol^{-1} , R is the gas constant, T is the temperature in Kelvin at which the equilibrium is established and K_c is the equilibrium constant.

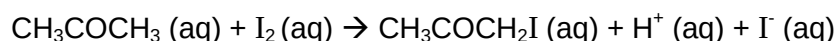
When 1.00 g of propanone was dissolved in 25 cm^3 of water and the solution allowed to reach equilibrium, the concentration of the 1,1-diol was found to be $0.0497 \text{ mol dm}^{-3}$.

Use the data given to calculate K_c and hence $\Delta G^\ominus$ of the reaction. You may assume that $[\text{H}_2\text{O}]$ is 55.5 mol dm^{-3} throughout.

- (iii) Comment and explain on the sign of $\Delta G^\ominus$ with reference to relative stability of the reactants to product.

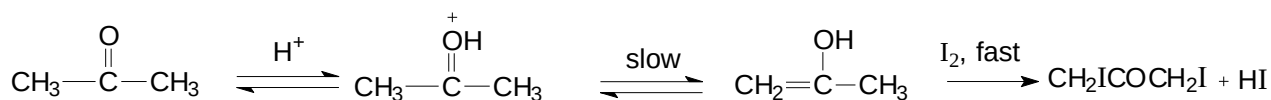
[9]

- (b) Propanone reacts with iodine in the presence of an acid catalyst.



The rate of reaction was studied via a colorimeter method, in which the colour of iodine was measured at regular time intervals.

The mechanism of this reaction is thought to be as follows:



- (i) The rate of reaction is directly proportional to $[\text{H}^+]$. Using this information and the mechanism above, write the rate equation for this reaction.
- (ii) Hence, sketch the graph of $[\text{I}_2]$ against time that you would expect to obtain.

- (iii) In the experiment, the following concentrations of reactants were used and the half-life of propanone was 4 min.

Propanone: 4.00 mol dm^{-3} , Iodine: $0.005 \text{ mol dm}^{-3}$, H^+ : 1.00 mol dm^{-3}

Predict and explain the half-life of propanone if the concentration of H^+ used was 2.00 mol dm^{-3} instead.

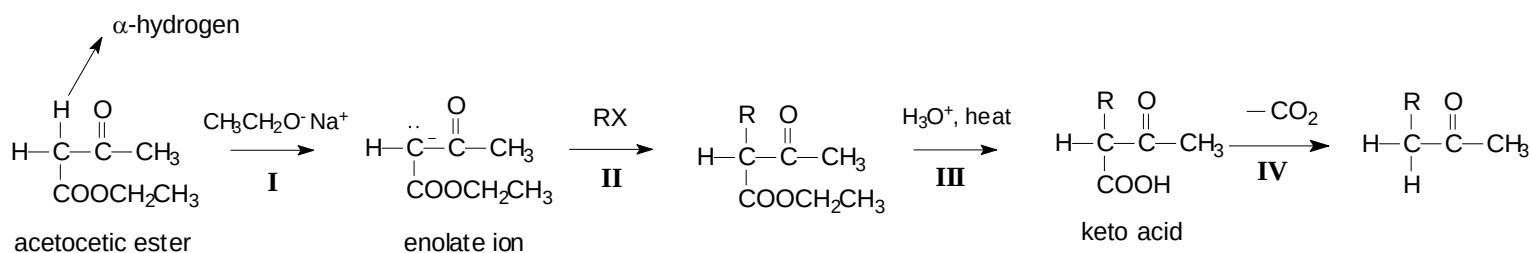
- (iv) Suggest another method, other than colorimetric method, that can be used to study the rate of this reaction.

[5]

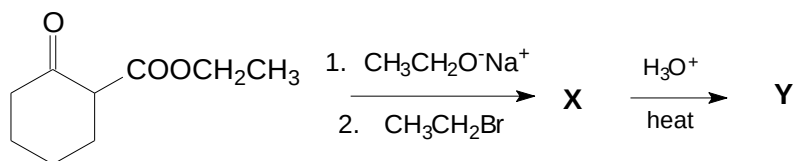
- (c) Propanone is a methyl ketone. Longer chain methyl ketones can be prepared from halogenoalkanes by the acetoacetic ester synthesis.

Acetoacetic ester, can be converted into its enolate ion, which is then alkylated by reaction with a halogenoalkane.

On heating the alkylated acetoacetic ester with aqueous HCl , a keto acid is formed which will then lose CO_2 to yield the ketone product.



- (i) In step I, the α -hydrogen atom of the ester can easily be removed by relatively mild bases like sodium ethoxide.
Suggest a reason why the α -hydrogen atom of the ester is unusually acidic.
- (ii) State the type of reaction for steps II and III.
- (iii) Deduce the structure of the halogenoalkane that can be used to prepare 5-phenylpenta-2-one using an acetoacetic ester synthesis.
- (iv) The following keto ester can also undergo the same sequence of reactions as acetoacetic ester.



Suggest the structures for X and Y.

[6]

[Total: 20]

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